

A Deterministic Least Fixed-Point Certificate for Spike Preservation in Trained Spiking Neural Networks under Reset-by-Subtraction Dynamics

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Abstract. Spiking neural networks (SNNs) are a leading paradigm for energy-efficient inference, but deterministic robustness guarantees for them remain scarce because the membrane *reset* couples binary spike decisions across time. The realistic reset-by-subtraction rule is the central obstacle: a single noise-induced spike flip alters the post-spike subtraction at the following step, which can flip a further spike, producing a temporal cascade that the standard per-step margin argument cannot bound. We introduce a deterministic, per-neuron, per-timestep certificate of spike preservation for a trained, direct-encoded leaky integrate-and-fire (LIF) layer under ℓ_2 -bounded input perturbation. The certificate replaces the unknown flip sequence with a conservative risk-flag sequence and solves the resulting self-referential bound as the *least fixed point* of a monotone operator on the Boolean lattice $\{0, 1\}^T$; existence follows from the Knaster–Tarski theorem and soundness from a causal temporal induction. The bound is *attack-agnostic* by construction. Across 27 trained configurations spanning three datasets, three leak factors, and three horizons, and a six-point perturbation sweep under both Gaussian and adversarial (FGSM) attacks, the certificate records *zero* soundness violations at every operating point while certifying 85–93% of hidden activations safe at small radius and degrading gracefully thereafter. We further show the guarantee composes soundly across depth: in a four-layer convolutional SNN with up to 98.8% clean accuracy, the certificate records zero violations at every layer. We position the result against analyses that respectively discard the reset term [4] or treat reset-by-subtraction only for random, non-leaky networks with average-case probabilistic bounds [1].

1 Introduction

Spiking neural networks (SNNs) compute with sparse, binary, asynchronous events and are the native model for neuromorphic hardware such as Loihi and SpiNNaker [3, 12, 17]. As SNNs move toward reliability-sensitive applications, a basic question follows the trajectory already charted for conventional networks: if the input is perturbed slightly, is the network’s computation preserved? For artificial neural networks (ANNs) this has produced a mature theory of certified robustness, spanning randomized smoothing [2] and formal verification via abstract interpretation, bound propagation, and reachability [6, 20, 22]. For SNNs the corresponding theory is far less developed, and the reason is specific and structural: the membrane reset.

The difficulty, intuitively. For a *single* timestep the question is elementary. If a neuron’s membrane sits at distance m from the firing threshold and the worst-case perturbation can displace it by at most B , then $m > B$ implies the spike cannot flip—one inequality settles it. The difficulty is entirely *temporal*. Under reset-by-subtraction, a neuron that fires has the threshold subtracted from its membrane, so whether it fired at $t - 1$ changes its membrane at t . If a perturbation flips a spike at some step, the corresponding subtraction is present in one trajectory and absent in the other, injecting a discrepancy that propagates forward and can flip further spikes. A single flip can cascade (Figure 1). The per-step margin argument breaks because the effective perturbation each neuron experiences at time t depends on *which*

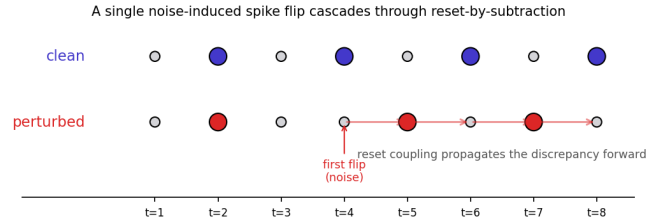


Figure 1: The core obstacle. A noise-induced flip at $t=3$ changes the post-spike subtraction, so the clean and perturbed trajectories diverge; the discrepancy propagates forward through the reset coupling. The certified bound at time t must therefore account for all earlier flips, yielding a self-referential dependency.

earlier spikes flipped—precisely the quantities one is trying to certify. This self-reference is the technical heart of the problem, and resolving it deterministically is our contribution.

The gap. Two recent analyses address SNN stability but each sidesteps this cascade. Ding et al. [4] derive a membrane-potential perturbation L_2 stability bound and convert it into a training regularizer; however, their derivation adopts the *reset-to-zero* rule and explicitly discards the reset fluctuation to obtain a smooth recursion, and their guarantee concerns the aggregate last-layer perturbation rather than individual spikes. Araya et al. [1] study noise sensitivity of LIF classifiers through Boolean (Fourier–Walsh) analysis [16] and *do* treat reset-by-subtraction, but restrict to randomly initialized, non-

leaky ($\beta=1$) networks and obtain *average-case probabilistic* bounds. Empirical defenses—adversarial and crafted-noise training [10, 18], input-boundary relaxation [11], and rate-coding randomized smoothing [14]—improve robustness but provide no deterministic, per-instance guarantee for the realistic dynamics. None delivers a deterministic, per-instance certificate for a *trained, leaky* network under *reset-by-subtraction*. This is the gap we fill.

Contributions.

1. A tight per-neuron displacement bound for the non-resetting case, improving on a global spectral bound by row-norm and partial-sum refinements (§3).
2. A reset-corrected bound cast as a monotone operator on $\{0, 1\}^T$, with existence of a least fixed point via Knaster–Tarski [9, 19] and a soundness proof by causal temporal induction; an iterative algorithm with $O(NT^2)$ cost per sample (§4–§6).
3. Empirical validation on trained MNIST, Fashion-MNIST, and CIFAR-10 classifiers: zero soundness violations across 27 configurations and a six-point ε -sweep under both random and adversarial attacks, certifying 85–93% of activations safe at small radius (§7).
4. A demonstration that the certificate *composes soundly across depth*: in a four-layer convolutional SNN with up to 98.8% clean accuracy, zero violations at every layer (§8).

2 Preliminaries

Discrete-time LIF neuron. We use the discrete leaky integrate-and-fire model [5, 7] with direct (constant-current) encoding: the flattened input $\mathbf{x} \in \mathbb{R}^d$ is injected at every timestep, so only the membrane evolves in time. For neuron $i \in [N]$ at timestep $t \in [T]$,

$$V(i, t) = \beta V(i, t-1) + (W_1 \mathbf{x})_i - \theta s(i, t-1), \quad (1)$$

$$s(i, t) = \mathbf{1}[V(i, t) \geq \theta], \quad V(i, 0) = 0, \quad (2)$$

with leak $\beta \in (0, 1)$, first-layer weights $W_1 \in \mathbb{R}^{N \times d}$ (rows $\mathbf{w}_i$), threshold θ (set to 1), N neurons, and horizon T . The term $-\theta s(i, t-1)$ is the subtractive reset.

Threat model. Given a clean input $\mathbf{x}$, an adversary chooses $\boldsymbol{\delta} \in \mathbb{R}^d$ with $\|\boldsymbol{\delta}\|_2 \leq \varepsilon$ and presents $\mathbf{x}' = \mathbf{x} + \boldsymbol{\delta}$. We write $V'(i, t)$, $s'(i, t)$ for the perturbed trajectory, $\Delta V(i, t) = V'(i, t) - V(i, t)$ for the membrane displacement, and $e(i, t) = |s'(i, t) - s(i, t)| \in \{0, 1\}$ for the absolute spike discrepancy.

3 The Clean-Case Bound

We first analyze the idealized non-resetting recurrence $V(i, t) = \beta V(i, t-1) + (W_1 \mathbf{x})_i$, where a closed-form bound is attainable.

Lemma 1 (Non-resetting displacement). *Under the non-resetting recurrence, for all i, t and all $\|\boldsymbol{\delta}\|_2 \leq \varepsilon$,*

$$|\Delta V(i, t)| \leq B_{\text{input}}(i, t, \varepsilon) := \varepsilon \|\mathbf{w}_i\|_2 \frac{1 - \beta^t}{1 - \beta}. \quad (3)$$

Proof. Unrolling from t to 1 with $V(i, 0) = 0$ gives $V(i, t) = \sum_{k=1}^t \beta^{t-k} (W_1 \mathbf{x})_i$, and identically for $\mathbf{x}'$, so $\Delta V(i, t) = \sum_{k=1}^t \beta^{t-k} (W_1 \boldsymbol{\delta})_i$. Since $(W_1 \boldsymbol{\delta})_i$ is independent of k , $|\Delta V(i, t)| = |(W_1 \boldsymbol{\delta})_i| \sum_{k=1}^t \beta^{t-k}$. By Cauchy–Schwarz, $|(W_1 \boldsymbol{\delta})_i| \leq \|\mathbf{w}_i\|_2 \varepsilon$, and $\sum_{k=1}^t \beta^{t-k} = (1 - \beta^t)/(1 - \beta)$. Multiplying gives (3). $\square$

Remark 1 (Tightness). *Bound (3) improves on the global spectral bound $\varepsilon \|W_1\|_2/(1 - \beta)$ on both factors: $\|\mathbf{w}_i\|_2 \leq \|W_1\|_2$, and the partial sum is strictly below $1/(1 - \beta)$ for finite t . The Cauchy–Schwarz step is tight in the worst case ($\boldsymbol{\delta} \parallel \mathbf{w}_i$), so (3) is the tightest row-wise linear bound.*

Definition 1 (Margin and flag). *The margin is $m(i, t) = |V(i, t) - \theta|$, and the flag is $\tau(i, t) = \mathbf{1}[m(i, t) < B(i, t, \varepsilon)]$, with $\tau = 1$ “at risk” and $\tau = 0$ “certified safe”.*

4 The Reset Obstacle and the Fixed Point

Retaining the reset term and subtracting the two instances of (1), $\Delta V(i, t) = \beta \Delta V(i, t-1) + (W_1 \boldsymbol{\delta})_i - \theta(s' - s)(i, t-1)$. Unrolling and applying the triangle inequality, using $|(s' - s)(i, k-1)| = e(i, k-1)$, gives the exact reset bound

$$|\Delta V(i, t)| \leq B_{\text{input}}(i, t, \varepsilon) + \theta \sum_{k=1}^t \beta^{t-k} e(i, k-1). \quad (4)$$

The second term contains the unknown flips e , the very quantities the perturbation alters. We replace $e(i, \cdot)$ by a risk-flag vector $\boldsymbol{\tau}_i \in \{0, 1\}^T$ and define the component-wise operator $\mathcal{F} : \{0, 1\}^T \rightarrow \{0, 1\}^T$,

$$\mathcal{F}(\boldsymbol{\tau}_i)_t = \mathbf{1} \left[m(i, t) \leq \underbrace{B_{\text{input}}(i, t, \varepsilon) + \theta \sum_{k=1}^t \beta^{t-k} \tau(i, k-1)}_{=: B_{\text{reset}}(\boldsymbol{\tau})_t} \right]. \quad (5)$$

The non-strict “ $\leq$ ” is deliberate; it is the hinge of the soundness proof.

Lemma 2 (Monotonicity). $\boldsymbol{\tau}_A \leq \boldsymbol{\tau}_B \Rightarrow \mathcal{F}(\boldsymbol{\tau}_A) \leq \mathcal{F}(\boldsymbol{\tau}_B)$ under the component-wise order on $\{0, 1\}^T$.

Proof. Every coefficient $\theta \beta^{t-k} \geq 0$, so $B_{\text{reset}}(\boldsymbol{\tau})_t$ is non-decreasing in each $\tau(i, k-1)$; the indicator $u \mapsto \mathbf{1}[m(i, t) \leq u]$ is non-decreasing in u ; the composition is order-preserving. $\square$

Since $(\{0, 1\}^T, \leq)$ is a finite complete lattice, Knaster–Tarski [19] guarantees a *least* fixed point $\boldsymbol{\tau}_{\min}^*$, obtained by the upward Kleene iteration $\boldsymbol{\tau}^{(0)} = \mathbf{0}$, $\boldsymbol{\tau}^{(n+1)} = \mathcal{F}(\boldsymbol{\tau}^{(n)})$, which converges in at most T steps. Among all fixed points it flags the fewest coordinates, hence certifies the most activations safe.

Algorithm 1 Least Fixed-Point Spike-Preservation Certificate

Require: clean run $V(i, t)$, weights $W_1, \beta, \theta, \varepsilon, T$

- 1: $m(i, t) \leftarrow |V(i, t) - \theta|$; $B_{\text{input}}(i, t) \leftarrow \varepsilon \|\mathbf{w}_i\|_2 (1 - \beta^t)/(1 - \beta)$
- 2: $\tau \leftarrow 0$
- 3: **repeat**
- 4: **for** $t = 1$ to T **do**
- 5: $r_t \leftarrow \theta \sum_{k=1}^t \beta^{t-k} \tau(i, k-1)$
- 6: $\tau_{\text{new}}(i, t) \leftarrow \mathbf{1}[m(i, t) \leq B_{\text{input}}(i, t) + r_t]$
- 7: **end for**
- 8: **until** $\tau_{\text{new}} = \tau$ ($\leq T$ iterations)
- 9: **return** $\tau_{\text{min}}^* \leftarrow \tau$

5 Soundness

Theorem 1 (Least fixed-point soundness). *For any fixed point τ^* of $\mathcal{F}$, $e(i, t) \leq \tau^*(i, t)$ for all t . Consequently $\tau^*(i, t) = 0 \Rightarrow e(i, t) = 0$: no certified-safe activation can flip, and the number of soundness violations is zero.*

Proof. Induction on t ; well posed because $B_{\text{reset}}(\tau^*)_t$ depends only on flags at $k < t$ (causality). *Base* ($t = 0$): $\Delta V(i, 0) = 0$, so $e(i, 0) = 0 \leq \tau^*(i, 0)$. *Step:* assume $e(i, k) \leq \tau^*(i, k)$ for $k < t$. Substituting into (4) and using non-negativity of coefficients, $|\Delta V(i, t)| \leq B_{\text{reset}}(\tau^*)_t$. If $\tau^*(i, t) = 0$, then since $\tau^* = \mathcal{F}(\tau^*)$ and (5) is non-strict, $\tau^*(i, t) = 0$ is the strict negation $m(i, t) > B_{\text{reset}}(\tau^*)_t$. Chaining, $|\Delta V(i, t)| \leq B_{\text{reset}}(\tau^*)_t < m(i, t)$, so $V'(i, t)$ cannot cross θ and $s'(i, t) = s(i, t)$, i.e. $e(i, t) = 0 \leq \tau^*(i, t)$. If $\tau^*(i, t) = 1$ the bound is immediate. The induction closes; specialize to τ_{min}^* . $\square$

Remark 2 (Attack-agnosticism). B_{input} depends only on $(\varepsilon, \|\mathbf{w}_i\|_2, \beta, \theta, T)$ and not on the direction of δ . The certified-safe set is therefore identical for every ℓ_2 -bounded perturbation, the worst case included; the certificate need not be recomputed per attack.

6 Algorithm and Complexity

Algorithm 1 computes τ_{min}^* by upward Kleene iteration. The inner reset-term sum dominates; maintaining the discounted partial sums incrementally and using that each coordinate flips at most once yields a per-sample cost of $O(NT^2)$ and memory $O(NT)$.

7 Experiments

Setup. A one-hidden-layer LIF classifier ($N=128$, reset-by-subtraction, direct encoding) with a linear readout on time-averaged spike counts, trained with Adam (10^{-3} , 3 epochs) and a triangular surrogate gradient [15, 21]. Perturbations are scaled to exact radius $\|\delta\|_2 = \varepsilon$: Gaussian, and FGSM [8]. We report **viol** (pairs with $\tau_{\text{min}}^*=0$ yet $e=1$; must be 0) and the **certified-safe fraction** $\text{mean}(1 - \tau_{\text{min}}^*)$, over 50 test samples per configuration.

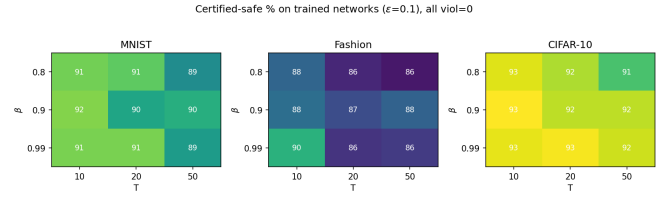


Figure 2: Certified-safe % on trained networks at $\varepsilon=0.1$ across leak β (rows) and horizon T (columns). All 27 cells satisfy $\text{viol} = 0$.

safe (%)	0.1	0.25	0.5	1.0	2.0	4.0
MNIST	90.4	86.7	81.0	68.5	48.7	31.6
Fashion	86.5	82.2	74.3	60.2	37.1	14.3
CIFAR-10	92.5	88.7	83.0	71.5	51.1	22.4

Table 1: Certified-safe fraction vs. ε ($\beta=0.9, T=20$). Every entry has $\text{viol} = 0$.

Soundness on the (β, T) grid. At $\varepsilon=0.1$, across all 27 configurations ($\beta \in \{0.8, 0.9, 0.99\}$, $T \in \{10, 20, 50\}$, three datasets), the certificate registers *zero* violations and certifies 85–93% of activations safe, only weakly dependent on β and T (Figure 2).

Degradation with radius. Fixing $\beta=0.9, T=20$ and sweeping $\varepsilon \in \{0.1, 0.25, 0.5, 1, 2, 4\}$, the certified-safe fraction decays smoothly and monotonically while soundness holds throughout (Table 1, Figure 3). At $\varepsilon=1.0$ the certificate still provably secures 60–72% of activations.

Random vs. adversarial. Replacing Gaussian noise by FGSM leaves the certified-safe fractions *identical* (an immediate consequence of attack-agnosticism, Remark 2) while the *actual* flip-rate rises 3–4 $\times$ at large ε (Figure 4). Crucially, $\text{viol} = 0$ holds at all 18 adversarial operating points, confirming the worst-case bound covers a directed attack.

8 Compositional Soundness Across Depth

The certificate is defined per spiking layer; we now show it *composes soundly* through a deep network. We train a four-layer convolutional SNN (three Conv-LIF layers and one FC-LIF layer, reset-by-subtraction throughout, batch-normalized, with a rate readout) to high accuracy (98.8% MNIST, 92.0% Fashion-MNIST, 70.0% CIFAR-10 at $T=8$). The certificate is applied layer by layer: layer 1 uses the input radius ε ; each subsequent layer uses the certified upstream flags to form its input bound, with batch-norm folded into the effective convolution weights. We evaluate viol and the certified-safe fraction at every layer on 300 test samples at $\varepsilon=0.1$.

Soundness composes. The decisive observation is that $\text{viol} = 0$ at every layer for all three trained datasets (Table 2, Figure 5). The per-spike guarantee of The-

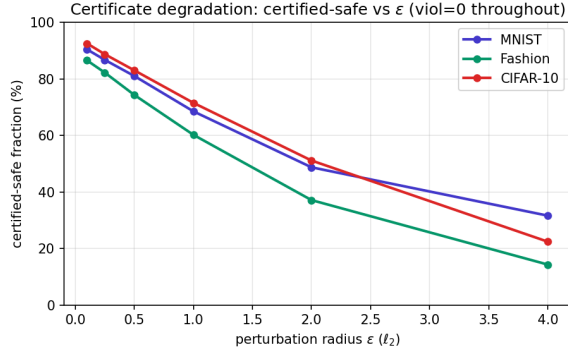


Figure 3: Certified-safe fraction vs. ε . Soundness holds at every radius; certifying power decays gracefully.

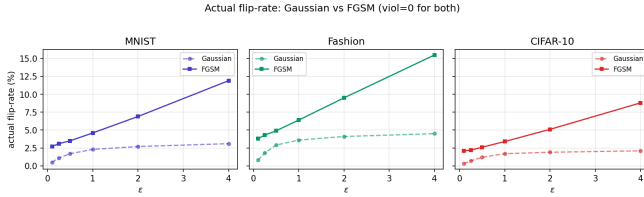


Figure 4: Actual flip-rate, Gaussian (dashed) vs. FGSM (solid). FGSM is the harder attack (3–4 $\times$ more flips at $\varepsilon=4$), yet the certificate records zero violations for both.

orem 1 is therefore preserved through composition: at no layer, on any of the 300 samples, did the certificate declare a spike safe that subsequently flipped. This validates the propagation construction as a sound primitive for deep spiking verification.

9 Related Work

SNN robustness. Empirical work shows SNNs possess partial inherent robustness depending on encoding and neuronal nonlinearity [18], improvable under crafted-noise or adversarial training [10]. Liang et al. [11] relax the spike nonlinearity to bound spike-input ranges, and Mukhoty et al. [14] transfer randomized smoothing [2] to rate-encoded SNNs for probabilistic certificates. These are complementary to our deterministic, per-instance guarantee.

Stability analyses. Ding et al. [4] analyze membrane-potential perturbation under reset-to-zero, discarding the reset term. Araya et al. [1] apply Boolean function analysis [16] to obtain average-case noise-sensitivity bounds for random, non-leaky networks under reset-by-subtraction. Moser et al. [13] study the integrate-and-fire neuron as a quantizer and identify the same reset term as the obstacle from a signal-processing vantage. Table 3 situates our contribution.

ANN verification. Deterministic certification for ANNs spans abstract interpretation [6], bound propagation [22], and star-set reachability [20]. Our operator-on-

	acc	L1	L2	L3	L4
MNIST	98.8	63.3	11.4	2.0	—
Fashion	92.0	59.3	8.3	0.9	—
CIFAR-10	70.0	59.6	3.4	0.1	—

Table 2: Certified-safe % by layer in the four-layer conv-SNN ($T=8$, $\varepsilon=0.1$, $n=300$). Clean accuracy (%) in column “acc”. *Every layer satisfies* viol = 0: soundness composes across depth. The certified fraction reduces with depth (§10).

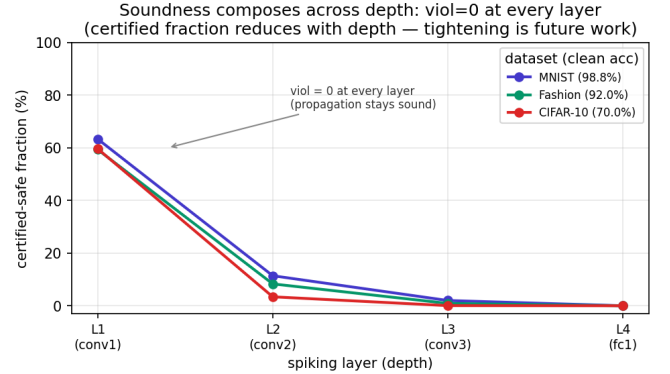


Figure 5: Certified-safe fraction by depth ($T=8$, trained networks). Soundness holds at every layer (viol = 0); the certified fraction reduces with depth as the worst-case bound accumulates, identifying tighter propagation as the natural next step.

a-lattice formulation is in the same deterministic spirit, specialized to the temporal reset coupling unique to SNNs.

10 Discussion, Limitations, Future Work

The central hypothesis—that the reset-induced flip cascade can be bounded deterministically by a self-referential risk-flag fixed point—is supported: $\tau_{\min}^*$ is provably sound (Theorem 1) and empirically unviolated across all single-layer operating points including under adversarial attack, while remaining non-vacuous (certifying the majority of activations at realistic radii). The guarantee further composes soundly through a four-layer network (§8).

Limitations and future work. The certificate is on individual spikes; lifting it to a non-vacuous *output-level* guarantee at depth requires tighter propagation, as the certified fraction reduces with depth (Table 2) under the worst-case row-norm step (Remark 1). Sharper propagation—e.g. symbolic bound propagation in the spirit of Zhang et al. [22], adapted soundly to the reset coupling—is the natural next step and the subject of ongoing work. A complementary direction is the reset geometry: Moser et al. [13] show reset-to-mod admits a clean quantization bound, which may inform a tighter per-layer primitive.

	Ding et al. [4]	Araya et al. [1]	This work
Reset	to-zero [†]	subtraction	subtraction
Weights	trained	random	trained
Bound	determ. L_2	prob., avg	determ., per-inst.
Leak β	uses	$\beta=1$	swept
Grain	layer	output (avg)	per neuron

Table 3: Positioning. [†] Ding et al. [4] further discard the reset term.

11 Conclusion

We introduced a deterministic, per-neuron certificate of spike preservation for trained LIF networks under reset-by-subtraction, formulated as the least fixed point of a monotone causal operator with existence by Knaster–Tarski and soundness by temporal induction. The certificate is attack-agnostic, and across 27 single-layer configurations and a six-point ε -sweep under both random and adversarial perturbations it produced zero violations while certifying the majority of activations safe at realistic radii; it further composes soundly across a four-layer convolutional network. The reset term that has obstructed deterministic SNN robustness analysis is handled not by discarding it nor by retreating to average-case bounds, but by absorbing it into a sound, computable fixed point on trained networks.

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